



Town of Brookline

Conservation Commission

Marcus Quigley, Chair
Roberta Schnoor, Vice Chair
Werner Lohe
Pallavi Kalita Mande
Pamela Harvey
Samuel Burrington
Will Corrdin

Associates
Marian Lazar

Brookline Conservation Commission

TUESDAY, MAY 30, 2023

7:00 PM – 7:50 PM

ZOOM WEBINAR

TO ATTEND VIRTUALLY, REGISTER IN ADVANCE FOR THIS WEBINAR:

[HTTPS://BROOKLINEMA.ZOOMGOV.COM/WEBINAR/REGISTER/WN_9LCv6FQ6RW2H0MREWEv3PQ](https://brooklinema.zoomgov.com/webinar/register/WN_9LCv6FQ6RW2H0MREWEv3PQ)

TO ACCESS THE MEETING BY PHONE, DIAL: US: +1 669 254 5252 or +1 646 828 7666

Webinar ID: 161 060 2001 Password: 75722251

MEETING FILES LINK

<https://www.brooklinema.gov/DocumentCenter/Index/4225>

AGENDA

- | | |
|----------------|--|
| 7:00 PM | Agenda Review/ Minutes |
| 7:05 PM | Continued Public Hearing of the Notice of Intent DEP 120-199 BWB 23-02
– 20 Newton Street |
| 7:30 PM | Climate Change Regulation Discussion |
| 7:40 PM | Staff update |
| 7:50 PM | Adjourn |

MEMORANDUM

Subject: Response to Conservation Comments

Project: **Dexter Southfield Cornish Field**
Project No. 21022.00

Date: 25 May 2023

To: Conservation Commission
c/o Mr. Thomas D. Brady
Conservation Agent
Town of Brookline
333 Washington Street
Brookline, MA 02445

By: Megan Buczynski, PE
Principal Civil Engineer

Distribution: Jason Beckett, Dexter Southfield
(via email) Robert L. Allen, Jr. Mark Novak, Activitas

Dear Commissioners:

This memorandum addresses items discussed at the Conservation Commission Hearing on 05/09/23 and the site walk on 05/17/23 regarding the Dexter Southfield School Cornish Field Renovation. As previously described, the proposed project includes the resurfacing of the existing synthetic turf field area, removal of impervious area within the field area (net decrease at the site), the small expansion of Cornish Field into currently maintained grass athletic space, the installation of future athletic pole bases at the eastern side of the field, and paving of the existing gravel drive from the edge of parking lot to the field entrance. A small portion of the existing field area is within the 150' buffer zone to the isolated wetland located to the west of the project site and as such the project is jurisdictional under the Brookline Wetland Bylaw.

As discussed in the Stormwater Memo provided with the submission, the proposed work does not change the drainage characteristics at the site. However, during the first hearing it was noted that while capacity of the existing drainage basin and isolated wetland is not a concern, erosion at the existing drainage basin to the west of the site which outlets to the existing isolated wetland has been identified. Mr. Brady noted a concern regarding the velocity of stormwater flow to the drainage basin from the upgradient swale area as well as the gravel road area. The School noted that limited maintenance has occurred at the drainage basin and swale area to address erosion and agrees that improved maintenance practices are needed at the site.

To address this concern, Activitas has developed a draft Operation and Maintenance Plan for the western campus stormwater infrastructure, see attached sheet C0.1. This plan details the proposed additions of gabion-style check dams within the swale area and modifications to the gravel drive and associated drainage swale leading down towards the drainage basin. In addition this provides information regarding the stormwater valves at the isolated wetland outlet and at the baseball field, as well as a description of the required maintenance practices for the various stormwater infrastructure elements. This plan is still in draft form as additional time is

required to complete the plan to ensure it will adequately address the erosion concerns at the site.

DXSF humbly requests that the Commission consider allowing the completion of this plan be a condition set forth in the Order of Conditions and that the Project Team can work with the Conservation Department for approval, which could be a requirement prior to issuance of a Certificate of Compliance (or limited COC if applicable) allowing usage of the field.

We are hopeful that this may be an amenable condition to the project to appropriately address the main concern raised during the hearing and site walk. We look forward to discussing this in further detail at the 5/30/23 hearing with the hope that the hearing may be closed, and a draft Order of Conditions considered.

If you have any questions or require additional information, please do not hesitate to contact me at (781) 355-7040 or meb@activitas.com.

Respectfully,

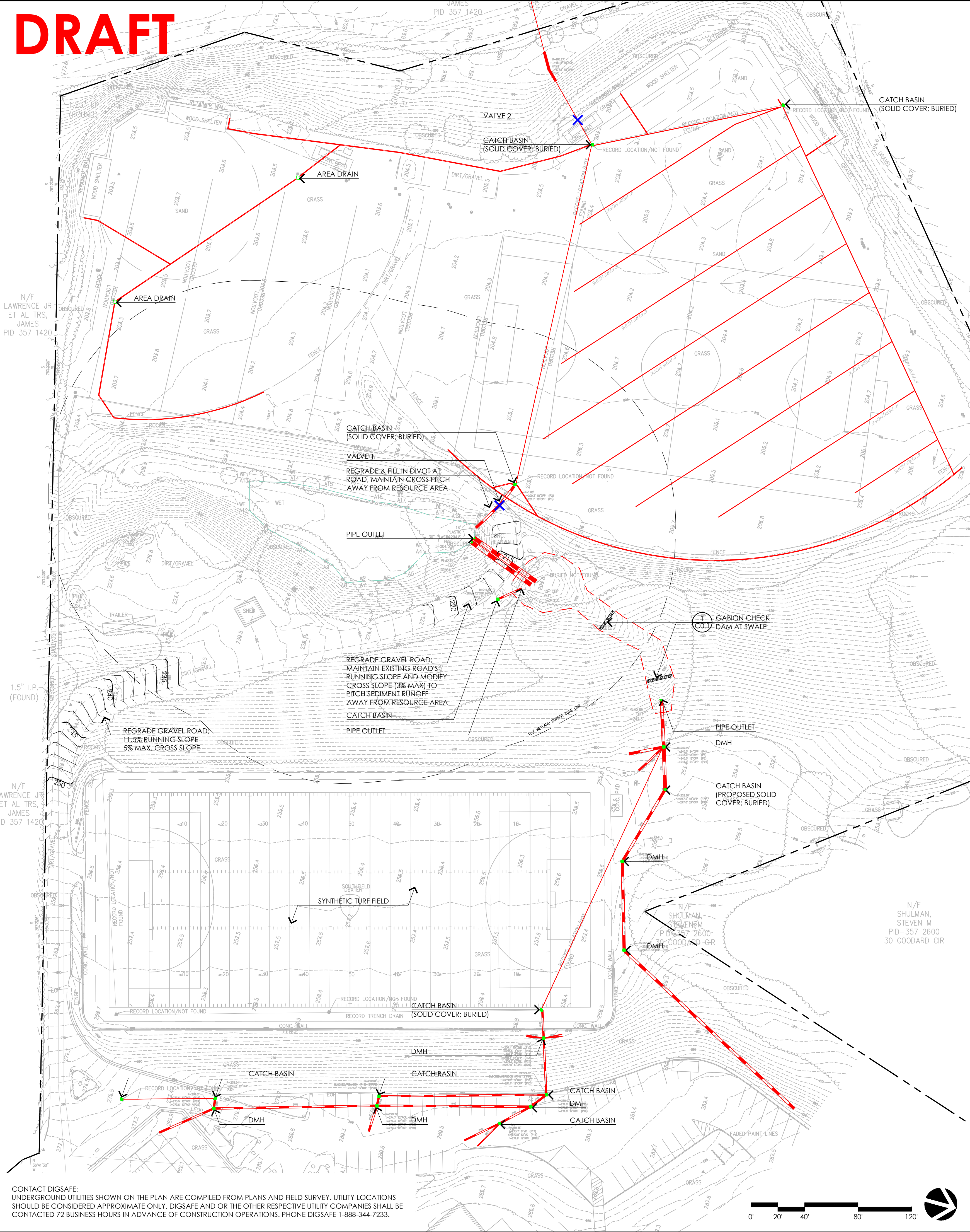
ACTIVITAS



Megan Buczynski, PE
Principal Civil Engineer
meb@activitas.com

Attachment: DRAFT O&M of Western Campus Stormwater – Sheet C0.1

DRAFT



STORMWATER VALVE OPERATION PROCEDURE

TWO (2) STORMWATER VALVES ARE INSTALLED AT THE DEXTER SOUTHFIELD BASEBALL/SOFTBALL STORMWATER SYSTEM TO TEMPORARILY REDUCE OR STOP STORMWATER FLOWING THROUGH THE DRAINAGE SYSTEM TO ALLENDALE FARM DURING OR FOLLOWING HEAVY STORMWATER FLOW CONDITIONS (E.G., IN THE EARLY SPRING MONTHS AND/OR FOLLOWING MULTIPLE STORM EVENTS). VALVE 1 IS AT THE OUTLET OF THE WETLAND BASIN AND THE VALVE 2 IS LOCATED BEHIND THE BASEBALL 1ST BASELINE BLEACHER AREA.

STORMWATER CAN BE HELD BACK TEMPORARILY ONLY BECAUSE THE DESIGN OF THE STORMWATER SYSTEMS REQUIRE RELEASE OF STORMWATER FROM THE SYSTEM. HOWEVER, IN THE CASE OF LARGE FLOW CONDITIONS HAVING A NEGATIVE EFFECT TO THE FARM AREA, IT IS RECOMMENDED THE FOLLOWING PROCEDURES ARE FOLLOWED:

- SIGNS WILL BE PLACED AT BOTH VALVE LOCATIONS SUMMARIZING THESE INSTRUCTIONS.
- THE FARM SHOULD CONTACT DEXTER'S FACILITIES DEPARTMENT AT (617) 522-5544 AND STATE THEY WOULD LIKE TO DISCUSS CLOSING THE STORMWATER VALVES TEMPORARILY AND ARE REQUESTING A MEETING ON-SITE TO REVIEW.
- REPRESENTATIVES FROM DEXTER'S FACILITIES DEPARTMENT AND FROM THE FARM SHOULD TOGETHER REVIEW:
 - THE ISSUES OCCURRING AT THE FARM DUE TO HEAVY STORMWATER FLOW.
 - THE FLOW OF STORMWATER FROM THE WETLAND TO CB-31 AND THE STORMWATER LEVEL WITHIN THE WETLAND AREA, IF THE WETLAND APPEARS TO BE AT CAPACITY, DISCUSSION ABOUT CLOSING THE VALVE IS REQUIRED. IF THERE APPEARS TO BE CAPACITY IN THE WETLAND AREA, THEN TEMPORARILY CLOSING THE VALVE MAY BE VIABLE.
 - THE WARNING TRACKS OF THE BASEBALL AND SOFTBALL FIELDS WHICH ALSO OPERATE AS THE INFILTRATION TRENCHES FOR THE SITE. IF THE WARNING TRACK AREAS ARE NOT TAKING IN WATER OR FEEL SPONGY UNDERFOOT, DISCUSSION ABOUT CLOSING THE VALVE IS REQUIRED. IF AT THE SURFACE THE WARNING TRACKS APPEAR TO BE DRAINING, THEN TEMPORARILY CLOSING THE VALVE MAY BE VIABLE.
- IF IT IS DETERMINED CLOSING THE VALVES IS VIABLE, THE DEXTER FACILITIES DEPARTMENT WILL CLOSE THE VALVES.
 - IMPORTANT NOTE: AT NO TIME SHOULD ONLY THE BASEBALL FIELD VALVE BE SHUT, AS THE POTENTIAL FOR STATIC PRESSURE FROM THE UPLAND SYSTEMS MAY NEGATIVELY AFFECT DRAINAGE INFRASTRUCTURE. THE OPTIONS FOR CLOSING ARE:
 - ONLY CLOSE THE WETLAND VALVE.
 - CLOSE BOTH THE WETLAND VALVE AND THE BASEBALL VALVE.
- FLOWING CLOSURE OF THE VALVE(S) DEXTER WILL MONITOR DAILY:
 - THE STORMWATER LEVEL WITHIN THE WETLAND BASIN TO CHECK THAT STORMWATER IS NOT BREACHING THE TOP OF THE BASIN.
 - WARNING TRACKS AT BASEBALL AND SOFTBALL TO ENSURE THAT HOLDING BACK THE STORMWATER IS NOT NEGATIVELY AFFECTING THE PLAYABILITY OF THE FIELD AREAS.
- THE FARM WILL MONITOR THE STORMWATER WITHIN THE FARM AREA AT THE STORMWATER SYSTEM OUTLET AND COMMUNICATE WITH DEXTER WHEN IT IS DETERMINED THAT STORMWATER RUNOFF CAN IDEALLY FLOW TO THE OUTLET AGAIN.

STORMWATER VALVE BACKGROUND INFORMATION

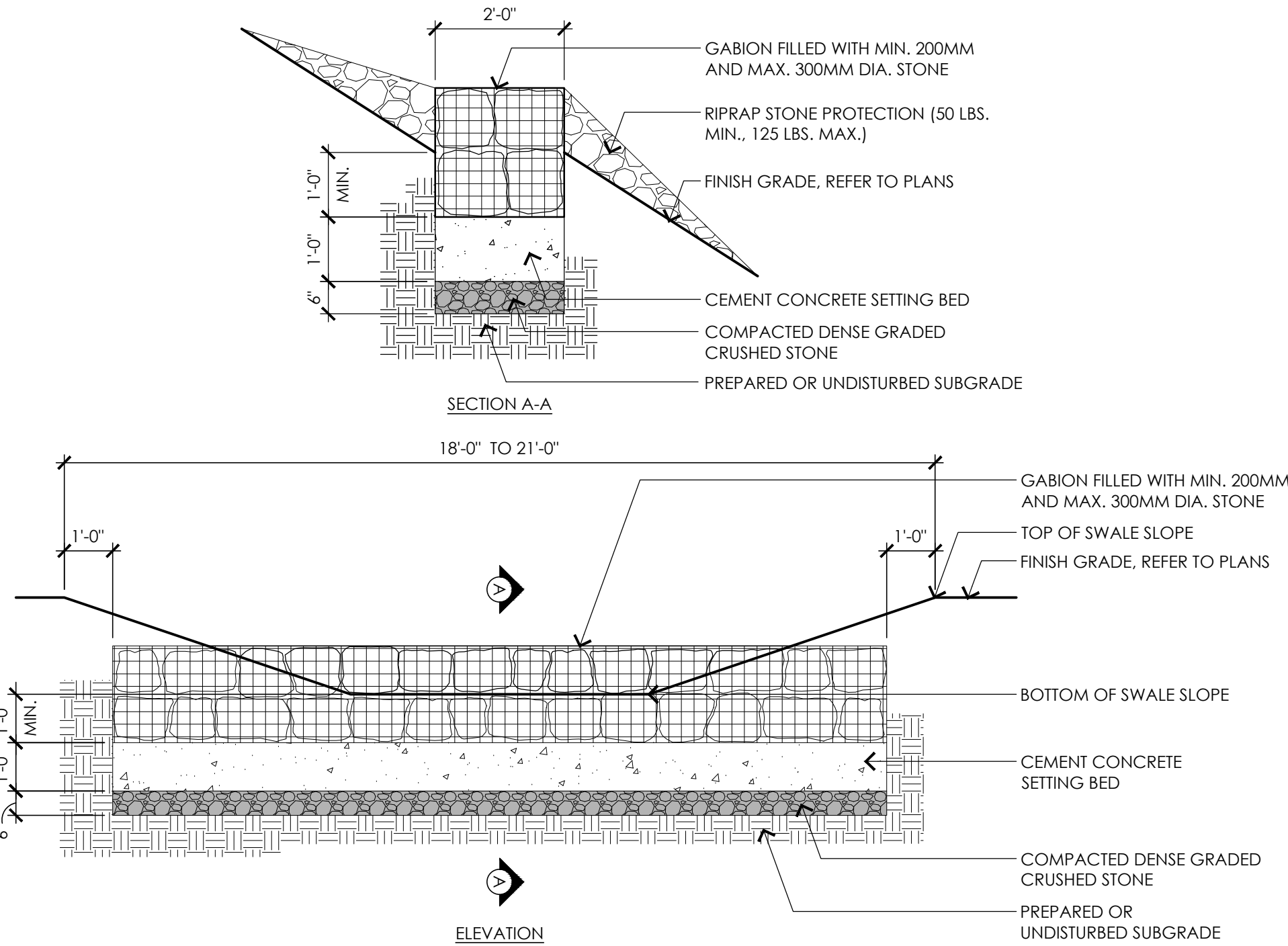
AS NOTED IN THE 8/28/2017 MEMORANDUM PROVIDED AS A FOLLOW-UP TO THE ORDER OF CONDITIONS FOR DEP FILE #120-165, BWB 17-06, THAT INCLUDED THE UPDATED GRADING AND UTILITY PLAN DATED 8/29/2017 (ATTACHED), THERE ARE CURRENTLY TWO (2) STORMWATER VALVES THAT CAN BE ADJUSTED TO OPEN OR CLOSED POSITIONS ADJACENT TO THE WETLAND BASIN (VALVE 1) AND THE BASEBALL FIELD AREA (VALVE 2) TO TEMPORARILY LIMIT/DECREASE STORMWATER FLOW TOWARDS ALLENDALE FARM.

VALVE 1 AT EXISTING CB-31
VALVE 1 IS LOCATED WITHIN EXISTING CATCH BASIN 31 (CB-31), TO THE EAST AND UP THE SLOPE FROM THE BASEBALL AND SOFTBALL FIELDS ON THE EDGE OF THE ACCESS DRIVE NEAR ELEVATION 219 AND IS A CONNECTION POINT FOR THE DRAIN LINE THAT PROVIDES OUTFLOW FOR THE EXISTING WETLAND BASIN LOCATED TO THE EAST OF THE BASEBALL AND SOFTBALL FIELDS. A LINE GATE IS INSTALLED ON THE OUTLET PIPE OF THIS CATCH BASIN, WHICH CAN BE OPEN OR SHUT USING A TOOL AT THE CATCH BASIN SURFACE. SHUTTING THIS VALVE STOPS STORMWATER FROM LEAVING THE WETLAND BASIN AND ANY STORMWATER THAT ENTERS CB-31 FROM THE SURFACE. IT SHOULD BE NOTED THAT DOCUMENTATION OF WHEN THIS VALVE WAS INSTALLED COULD NOT BE FOUND IN THE AVAILABLE INFORMATION FROM THE VARIOUS NOI/OOC FILINGS (1998, 2003, 2005). IT WAS INSTALLED PRIOR TO THE 2017 BASEBALL/SOFTBALL FIELD RENOVATION PROJECT.

VALVE 2 AFTER EXISTING CB-32
VALVE 2 IS LOCATED ON THE BACKSIDE OF THE BASEBALL 1ST BASELINE CONCRETE BLEACHER PAD. THIS VALVE IS IN LINE WITH THE 18" DRAIN LINE THAT CONNECTS EXISTING CATCH BASIN 32 (CB-32) TO THE CATCH BASIN LOCATED DOWN THE SLOPE OF THE ALLENDALE FARM PROPERTY. SHUTTING THIS VALVE STOPS STORMWATER THAT FLOWS THROUGH THE DRAINAGE NETWORK FROM THE WETLAND AREA (IF VALVE 1 IS OPEN); AND THAT IS COLLECTED IN THE INFILTRATION TRENCHES IN THE BASEBALL AND SOFTBALL OUTFIELD AND SIDELINE AREAS. IT SHOULD BE NOTED THAT THIS VALVE WAS INSTALLED AS PART OF THE 2017 BASEBALL/SOFTBALL FIELD RENOVATION PROJECT.

MAINTENANCE INSTRUCTIONS

STORMWATER ELEMENT	DESCRIPTION OF MAINTENANCE PRACTICE	YEARLY FREQUENCY
PIPE NETWORK	INSPECT AS NEEDED; IF A POTENTIAL CLOG IS IDENTIFIED - USE CAMERA TO REVIEW SYSTEM.	AS NEEDED
MANHOLES & CATCH BASINS	INSPECT UNITS 4X PER YEAR; CLEAN UNITS - WHEN DEPTH OF DEPOSITS IS GREATER THAN OR EQUAL TO ONE HALF DEPTH FROM THE BOTTOM OF THE INVERT OF THE LOWEST PIPE IN THE STRUCTURE.	4X PER YEAR
PIPE OUTLET LOCATIONS	INSPECT OUTLET 2X PER YEAR TO CHECK FOR EROSION AND MOVEMENT OF RIP RAP. REMOVE ANY DEBRIS/TRASH AT THE OUTLETS, REESTABLISH RIP RAP IF MOVEMENT HAS OCCURRED, RECOMPACT AREA OF EROSION AS NEEDED.	2X PER YEAR
STORMWATER SWALE & CHECK DAMS	INSPECT 2X PER YEAR AND FOLLOWING LARGE STORMS (OVER 1/2" OF RAINFALL) - REESTABLISH RIP RAP IF MOVEMENT HAS OCCURRED AND REVIEW FOR EROSION AND ADDRESS AS NEEDED.	2X PER YEAR
GRAVEL ROAD GRADING	INSPECT 1X PER YEAR MINIMUM AND FOLLOW A STORM OF 1/2" OF RAINFALL OR GREATER - REGRADE ROAD AS NEEDED AND CHECK FOR EROSION AT SIDE SLOPE AREAS AND RECOMPACT AND ADD RIP RAP AT EDGES AS NEEDED.	1X PER YEAR
SYNTHETIC TURF FIELD	REFER TO SPECIFIC TURF MAINTENANCE INSTRUCTIONS.	REFER TO SPECIFIC TURF MAINTENANCE INSTRUCTIONS
PARKING LOT SWEEPING	MECHANICAL SWEEPER.	2X PER YEAR
VALVE LOCATIONS	INSPECT 1X PER YEAR TO ENSURE WORKING PROPERLY.	1X PER YEAR



1 GABION CHECK DAM AT SWALE
NOT TO SCALE

CONSULTANTS

GEOTECHNICAL ENGINEER -
HALEY & ALDRICH

ELECTRICAL ENGINEER -
NV5

SURVEY -
VANASSE HANGEN BRUSTLIN

DEXTER SOUTHFIELD SCHOOL

BROOKLINE, MA

CORNISH FIELD RENOVATION

Stormwater Review
May 30, 2023

REVISIONS:

NO.	DATE	DESCRIPTION

SCALE: 1" = 40'-0"
PROJECT NO.: 21022.00
FILE: 21022.00-O&M_PLAN.dwg
DRAWN: KH
CHECKED: MEB

SEAL:

SHEET TITLE:

O&M OF WESTERN
CAMPUS STORMWATER

SHEET NO:

C0.1